Notation : In this sheet, the notation $a \lesssim b$ means that $a \leq C\, b$ where C is an absolute constant. For asymptotic inequalities that hold up to constant factors, we use the notation $a_N = O(b_N)$ which means that : $\exists\, C > 0$, $a_N \leq C b_N$ as $N \to \infty$. For two-sided inequalities, we write $a_N = \Theta(b_N)$ when $\exists\, c, C > 0$ such that $c b_N \leq a_N \leq C b_N$ as $N \to \infty$.

Some Classical Inequalities

Exercise 1. Let X be a r.v.. For $p > 0$, the p-th moment of X is defined as $\mathbb{E}\, X^p$, and the p-th absolute moment is $\mathbb{E}\, |X|^p$. This leads to the notion of L^p norm of X

$$\|X\|_p = (\mathbb{E}\, |X|^p)^{1/p}, \qquad p \in (0, \infty),$$

which can be extended to $p = \infty$ by the essential supremum of $|X|$, i.e. by $\|X\|_\infty = \text{ess sup}\, |X|$.

1. Recall the definition of essential supremum.
2. Show that $\|X\|_p$ is an increasing function in p, i.e. $\|X\|_p \leq \|X\|_q$ for any $0 < p \leq q = \infty$.
3. Recall Jensen, Cauchy-Schwarz, Hölder and Minkowski inequalities.

Exercise 2 (Expectation via tails ☕)**.** There is an important connection between the tails and the expectation of a random variable. The following identities are typically used to bound the expectation by the tails. Prove them.

1. *Integral identify* : If X is a non-negative r.v., then $\mathbb{E}\, X = \int_0^\infty \mathbb{P}\, [X > t]\ dt$.
2. *Generalization* : for any r.v. X, we have $\mathbb{E}\, X = \int_0^\infty \mathbb{P}\, [X > t]\ dt - \int_{-\infty}^0 \mathbb{P}\, [X < t]\ dt$.

More generally, the tails are connected to the moments of a random variable.

3. *p-th moment via the tail* : for any r.v. X and $p \in (0, \infty)$, we have $\mathbb{E}\, |X|^p = \int_0^\infty p\, t^{p-1}\, \mathbb{P}\, [|X| > t]\ dt$.

Exercise 3 (Tails via expectation and variance ☕)**.** For any r.v. X, prove the following.

1. *Markov's inequality* : If X is non-negative, then $\mathbb{P}\, [X \geq t] \leq \frac{\mathbb{E}X}{t}$ for any $t > 0$.
2. *Chebyshev's inequality* : If X has mean μ and variance σ^2, then $\mathbb{P}\, [|X - \mu| \geq t] \leq \frac{\sigma^2}{t^2}$ for any $t > 0$.

Limit Theorems

Exercise 4 (Limit theorems ☕)**.** Let $X_1, X_2, \ldots$ be a sequence of i.i.d. r.v. with mean μ and variance σ^2. Consider the sum $S_N = X_1 + \ldots + X_N$.

1. Compute the variance of the sample mean S_N/N. Interpret the result.
2. The law of large numbers makes this interpretation precise. Recall the strong LLN.
3. Recall the central limit theorem (CLT).
4. Show that $\mathbb{E}|\frac{S_N}{N} - \mu| \lesssim \frac{1}{\sqrt{N}}$ for any $N \geq 1$.
5. Show that $\mathbb{E}|\frac{S_N}{N} - \mu| = \Theta\left(\frac{1}{\sqrt{N}}\right)$ as $N \to \infty$.

Exercise 5 (Tails of the normal distribution ☕)**.** Let $g \sim N(0,1)$. Prove the following.

1. For all $t > 0$, we have

$$\left(\frac{1}{t} - \frac{1}{t^3}\right) \frac{1}{\sqrt{2\pi}} e^{-t^2/2} \quad \leq \quad \mathbb{P}[g \geq t] \quad \leq \quad \frac{1}{t} \frac{1}{\sqrt{2\pi}} e^{-t^2/2}.$$

In particular, for $t \geq 1$ the tail is bounded by the density :

$$\mathbb{P}[g \geq t] \ \leq \ \frac{1}{\sqrt{2\pi}} e^{-t^2/2}.$$

Exercise 6 (Special case of binomial distribution ☕)**.** Let $X_1, \dots, X_N \sim \mathrm{Ber}(p)$ Bernoulli r.v. with some fixed parameter $p \in (0,1)$.

1. What is the distribution of S_N ? the limit distribution of a (properly scaled) sum of X_i ?

Now suppose that $X_{i,N} \sim \mathrm{Ber}(p_N)$, with parameter p_N that decays to zero as $N \to \infty$ so fast that S_N has mean $O(1)$. The CLT fails in this regime. Instead, the following result says that S_N converges to a Poisson distribution.

Theorem 1 (Poisson limit theorem)**.** Let $X_{i,N}$, $1 \leq i \leq N$, be independent r.v. $X_{i,N} \sim \mathrm{Ber}(p_N)$, and let $S_N = \sum_{i=1}^N X_{i,N}$. Assume that $Np_N \to \lambda < \infty$. Then, as $N \to \infty$,

$$S_N \to \mathrm{Pois}(\lambda) \qquad \text{in distribution.}$$

2. Prove the Poisson limit theorem.

Application

Exercise 7 (Binomial tails ☕☕)**.** Toss a fair coin N times. Let p_N be the probability that we get at least $3N/4$ heads.

1. Show that $p_N \lesssim 1/N$.
2. Can we hope a faster decay (than $1/N$) ? At least asymptotically (as $N \to \infty$) ?